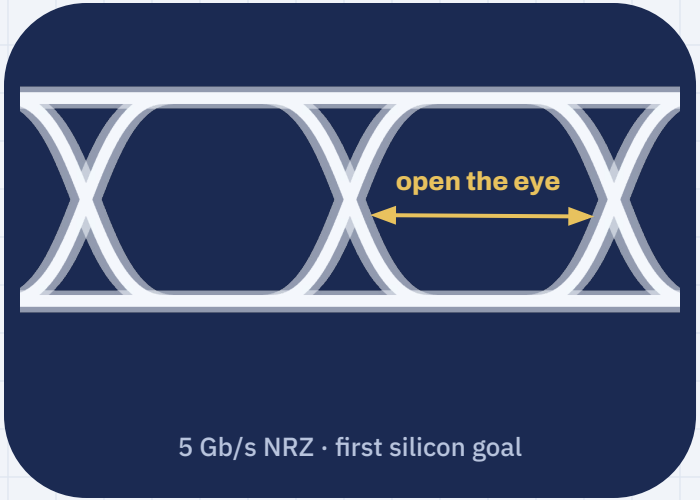
IHP SG13G2 · 130 NM SIGE BICMOS · OPEN SILICON

Open-SERDES

Towards an open-source high-speed serial link

Every fast interface rides on a SerDes, yet SerDes IP is closed. **Open-SERDES** is an **open project** to build a high-speed serial transceiver and a **reproducible channel-to-GDS flow** on open tools, for anyone to study, reuse and extend.



[GitHub repo link]

SG13G2
IHP open PDK

FFE · CTLE
equalization

CDR
clock & data recovery

Ch → GDS
open design flow

Biomedical Research in Integrated Circuit Design
Where biomedical engineering meets IC design innovation

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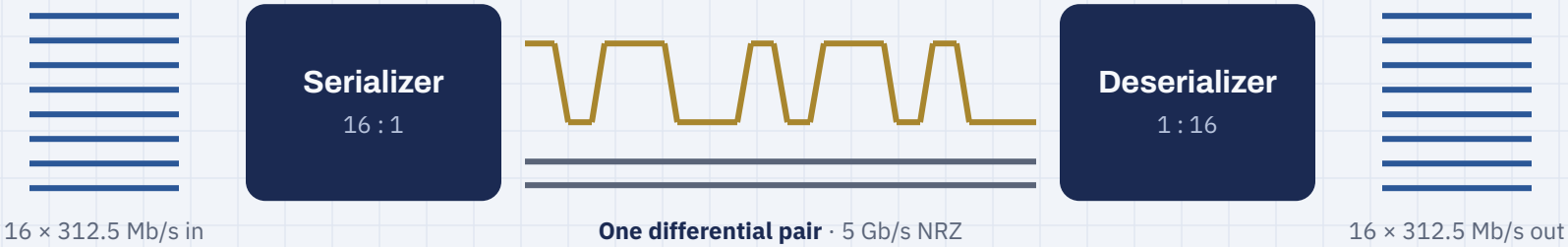
Raise your hand if…

- You used a SerDes today (USB, PCIe, Ethernet)
- You have looked at an eye diagram
- You have designed a CML gate or a StrongARM latch
- You have simulated a channel from S-parameters
- You have taped out a chip on an open PDK
- **You know an open SerDes you could tape out today**

That last hand is why we're here.

Open PDKs opened up digital and analog design. The high-speed link is still a black box.

Every fast interface is a SerDes



16 × 312.5 Mb/s in

One differential pair · 5 Gb/s NRZ

16 × 312.5 Mb/s out

Fewer pins, no skew
One differential pair with an embedded clock replaces a wide, skew-limited bus.

Everywhere
PCIe, USB, Ethernet, SATA, DisplayPort and chiplet links all ride on SerDes.

Still closed
Production SerDes IP is proprietary, NDA-bound and locked to one process.

What exists in the open today

Effort	What it is	Speed	What's missing
OpenSerDes (Purdue, DATE 2021)	All-digital link in SKY130 via OpenLANE; inverter TX/RX, oversampling CDR	2 Gb/s, simulated	Equalization; CMOS-speed-limited
serdespy · PyBERT · JLSD	Open Python / Julia link models: channels, FFE, CTLE, DFE, eyes	Behavioral	No tie to a PDK or real circuits
IIC-OSIC-TOOLS + SG13G2 AMS template (JKU)	Open tool container and Makefile flow to a tapeout-ready AMS chip	n/a	No high-speed link IP
Commercial SerDes IP	Silicon-proven transceivers from IP vendors and foundries	100+ Gb/s per lane	Closed, NDA, per-process licence

The gap: as far as we know, no open PDK yet has a multi-Gb/s mixed-signal transceiver with EQ and CDR, or a reproducible channel-to-GDS flow for one.

PART 1

Why IHP SG13G2?



SG13G2: a BiCMOS process built for speed

300 GHz
SiGe:C HBT transit frequency (fT)

500 GHz
HBT max. oscillation frequency (fmax)

7 metals
incl. 2 µm and 3 µm thick top Al layers

1.2 V
core CMOS, plus 3.3 V thick-oxide devices

WHAT THAT BUYS A SERIAL LINK

HBT CML with headroom
fT far above the 2.5 GHz Nyquist rate: fast muxes, drivers and latches.

Inductors on thick metal
Shunt-peaked loads, LC-VCOs and low-loss lines on TopMetal1 and TopMetal2.

Analog-friendly passives
MIM caps, poly resistors and varactors for CTLEs, terminations and loop filters.

A real path to silicon
IHP open-source MPW runs and Tiny Tapeout IHP shuttles.

Source: IHP Open-PDK documentation, SG13G2 process specs

We don’t start from zero

OURS · SMACD 2026

2.4 GHz LC-VCO fractional-N PLL

- Fully open-source flow in SG13G2
- 4 nH spiral, Q = 16.8 (openEMS)
- −100.8 dBc/Hz at 1 MHz offset
- 12.73 mW · 0.347 mm²

UOM PROJECT · 2016

5 Gb/s SerDes model in 28 nm

- Commercial 28 nm generic library
- Fast-locking ring-VCO PLL
- Under 10 ps peak-to-peak jitter
- Architecture insight at 5 Gb/s

PURDUE · DATE 2021

OpenSerDes in SKY130

- All-digital, fully synthesizable
- OpenLANE RTL-to-GDS automation
- 2 Gb/s in post-layout simulation
- Proof that open automation works

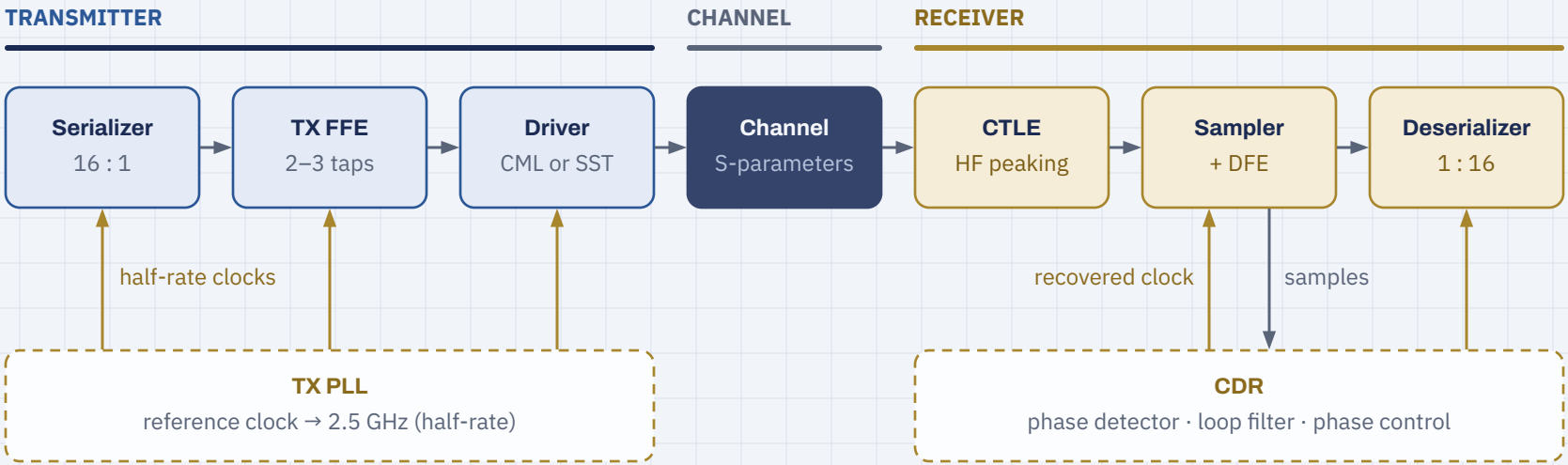
Open-SERDES = 5 Gb/s architecture insight + open automation + an open SG13G2 clock source we have already built.

PART 2

Anatomy of a Serial Link



Anatomy of a serial link



Grey arrows carry data, gold arrows carry clocks. Every box here is an open design choice; the next two slides size and scope them.

5 Gb/s, in numbers

200 ps
Unit interval: each bit lasts 1 / (5 Gb/s)

2.5 GHz
Nyquist frequency the channel must carry (NRZ)

10^{-12} BER
Typical bit-error-rate target for a serial link

Same rate class as **PCIe Gen 2**, **USB 3.2 Gen 1** and **5GBASE-KR** (IEEE 802.3cb).

TO BE SET BY THE FEASIBILITY STUDY

Channel loss
How much loss at 2.5 GHz must we equalize?
[__ dB]

Jitter
What total jitter can the PLL and CDR afford?
[__ ps pk-pk]

Power
Which energy per bit is competitive at 130 nm?
[__ pJ/bit]

Area
What silicon budget does one lane get?
[__ mm²]

The design space we'll explore

Modulation

NRZ PAM4

Start: **NRZ**, simplest receiver and best SNR at 5 Gb/s.

TX driver

CML (HBT) **SST voltage-mode**

Start: **HBT CML**; SST as the low-power follow-up.

Equalization

TX FFE **CTLE** **DFE**

Start: FFE + CTLE + 2-tap half-rate DFE; more taps only if the channel demands it.

Sampler

HBT CML latch **StrongARM**

Start: **build both** and compare speed, offset and power.

Clock rate

full **half** **quarter**

Start: **half-rate**, a 2.5 GHz clock next to our 2.4 GHz PLL.

CDR

bang-bang **baud-rate** **oversampled**

Start: **bang-bang PD** with a digital loop filter.

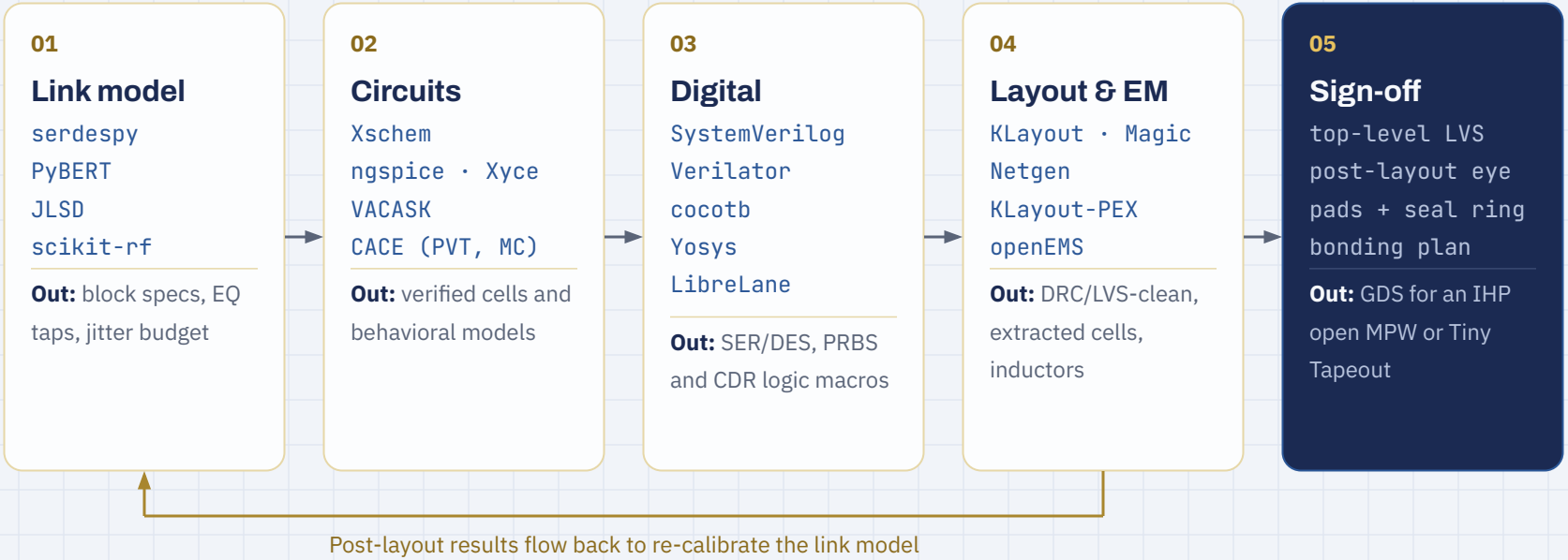
Filled pills: baseline candidates. The feasibility study makes the final call.

PART 3

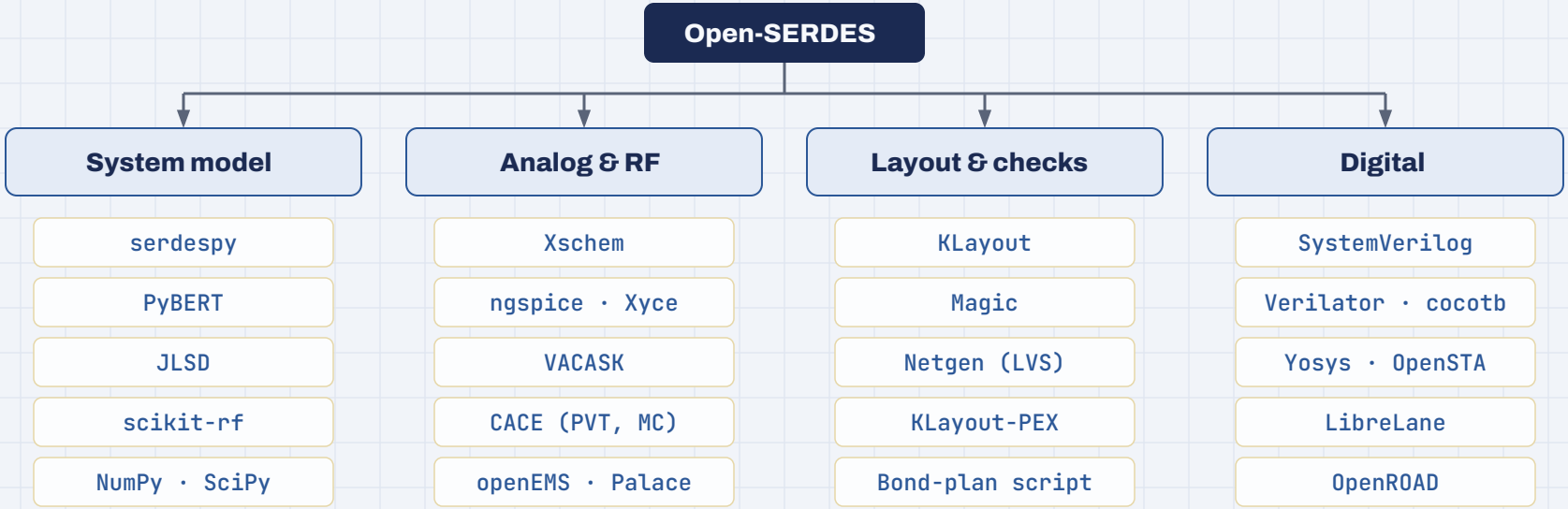
An Open Flow, Channel to GDS



The Open-SERDES flow



Tool map: every step on open tools



PDK: IHP SG13G2 Open-PDK · every tool ships in the IIC-OSIC-TOOLS container

Closing the loop: model ↔ silicon

SYSTEM

Link model

`serdespy · PyBERT · JLSD`

Channel, FFE, CTLE, DFE, CDR and jitter at the bit level

Specs: gain, peaking, swing, jitter



Extracted: pulse response, CTLE H(f), noise

CIRCUIT

SG13G2 cells

`Xschem · ngspice · Xyce`

Transistor-level and post-layout, across PVT corners

- Statistical eye and BER for every design point
- A Python GUI to explore taps and settings live
- Every number in the model traceable to a simulated cell

One repository, one command

- Built on JKU's **ihp-sg13g2-ams-chip-template**: Makefile-driven, recursive macros
- Each block is a macro with schematic, layout, testbenches and verification
- One spec file shared by the link model and the testbenches
- GitHub Actions regression: DRC, LVS and eye-opening checks on every commit
- Open licence throughout [Apache 2.0 / Solderpad]

PROPOSED LAYOUT

```
open-serdes/
├ model/                    link model + GUI
├ channels/                 S-parameter files
├ macros/                   one folder per block
│  ├ tx_serializer/
│  ├ tx_ffe_driver/
│  ├ rx_ctle/
│  ├ rx_sampler_dfe/
│  ├ pll_lcvco/             from our SMACD PLL
│  └ cdr/
├ rtl/                      PRBS, SER/DES, CDR logic
├ verification/             regression + eye checks
└ Makefile
$ make model && make all
```

One command: the make targets



Why it’s hard, and why it’s worth it

THE HARD PARTS

Models at GHz
HBT compact models and parasitics must hold up at 2.5 GHz and beyond.

Open extraction and EM
RF-grade extraction of pads, ESD, inductors and lines is still maturing.

Jitter in simulation
PLL and CDR transients are slow, so we need calibrated behavioral models.

Packaging and test
Bondwires, a high-speed test PCB, and access to a BERT or a fast scope.

WHAT THE COMMUNITY GETS

- ✓ A first open reference design for multi-Gb/s links
- ✓ A reusable HBT / CML cell library for SG13G2
- ✓ A calibrated, open link model anyone can extend
- ✓ A teaching path from link theory to real silicon

PART 4

What We’ll Build & Test



Highlights & deliverables

HIGHLIGHTS

1. First open multi-Gb/s mixed-signal SerDes on an open PDK
2. 5 Gb/s NRZ transceiver: TX, RX, PLL and CDR in SG13G2
3. One reproducible flow from channel model to GDS
4. A link model calibrated against post-layout and silicon
5. Everything public: design files, models and lab data

DELIVERABLES

GDSII
tapeout-ready 5 Gb/s transceiver

RTL + TBs
SER/DES, PRBS, CDR logic and benches

Link model
Python simulator and GUI

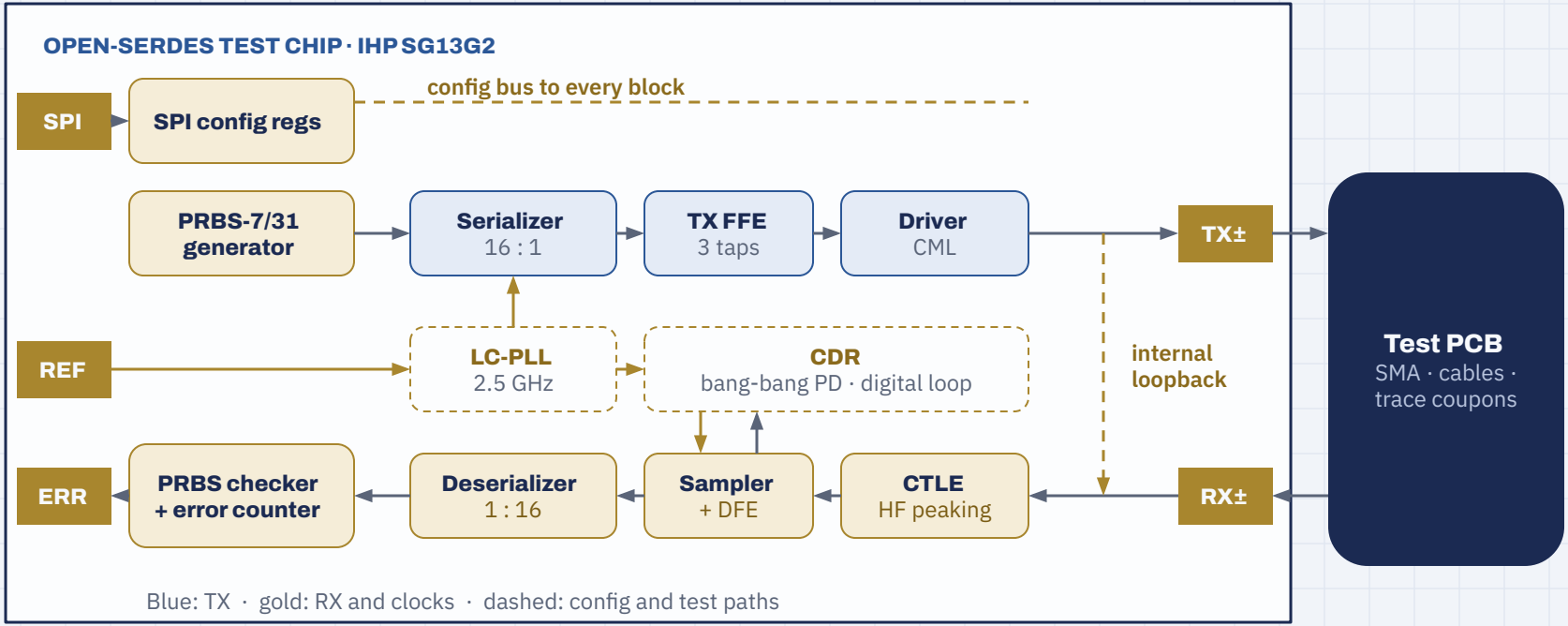
Cell library
HBT / CML cells for SG13G2

Docs
GitHub repo and tutorial

Paper + data
JSSC / TCAS, SMACD 2027, open data

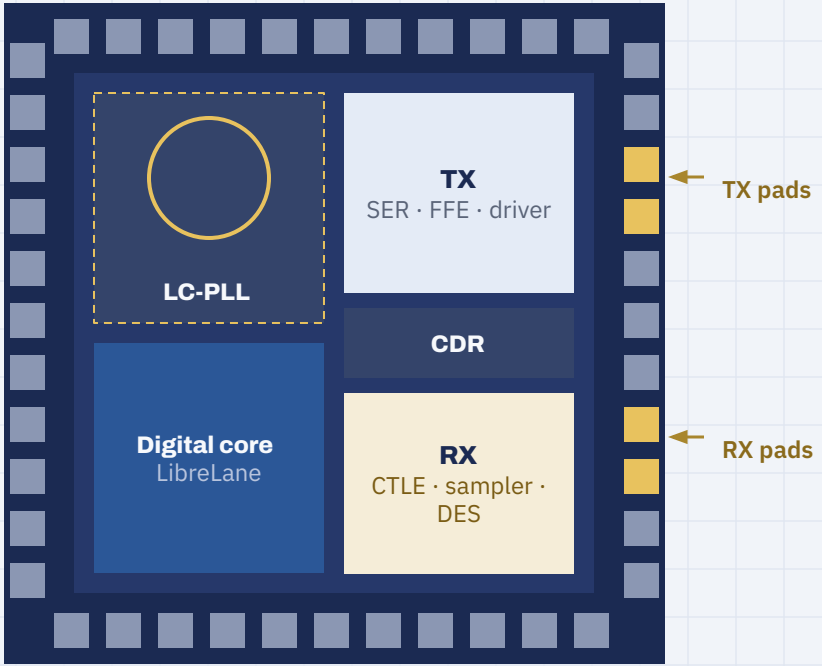
Goal: clone the repo, run make all, and reproduce the chip, the models and the measurements.

Chip block diagram (concept)



Floorplan concept

- Die of roughly [2 × 2] mm² in SG13G2 (to be sized)
- TX and RX pads on one edge, shortest bond wires
- LC-PLL in its own corner, guard ring, own supply
- Separate AVDD, DVDD and VDDIO domains
- Digital core hardened with LibreLane
- Seal ring, fill and logo from the template flow



Simulation testbench & sign-off



Sweeps

- Channel loss: 0 to [__] dB at 2.5 GHz
- PVT corners and Monte Carlo
- Jitter and supply-noise injection
- FFE, CTLE and DFE settings

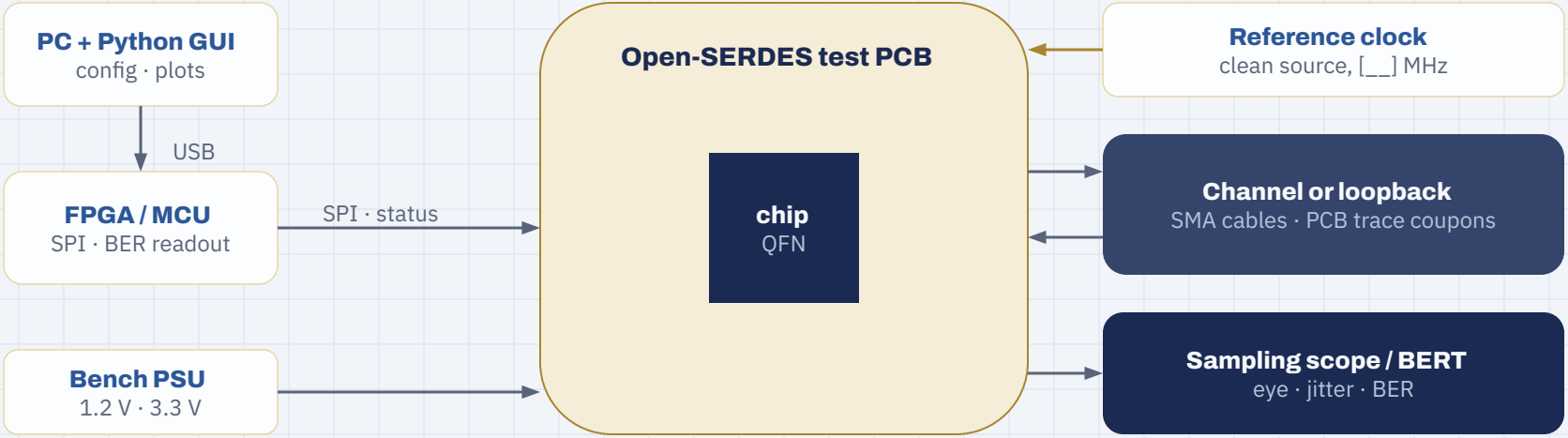
Pass criteria

- Eye height ≥ [__] mV
- Eye width ≥ [__] UI
- BER ≤ 1e-12 (statistical)
- Power ≤ [__] pJ/bit
- CDR lock in < [__] µs

Sign-off checks

- Full-chip DRC (KLayout)
- Full-chip LVS, no black boxes
- PEX and post-layout re-simulation
- CI regression on every commit

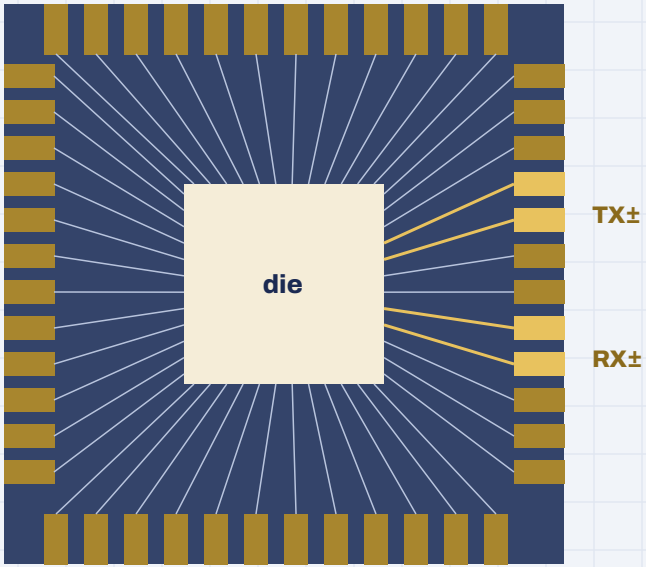
Test bench in the lab



WE WILL MEASURE

- Eye diagram
- BER bathtub
- Jitter (RJ / DJ)
- Power per bit
- CTLE response
- CDR lock time

Package & pinout (concept)



QFN-48, 7 × 7 mm (concept)

Group	Pins	Notes
High-speed	TXP, TXN, RXP, RXN	GSSG, shortest bond wires
Clock	REFCLK P / N	external reference
Control	SCK, MOSI, MISO, CS	SPI config + BER readout
Supplies	AVDD, DVDD, VDDIO, VSS	separate domains
Status	ERR, LOCK, DBG	debug and monitoring

Bonding plan generated from a config file, as in the JKU template flow.

PART 5

Roadmap & Community



Roadmap: blocks → 5 Gb/s → beyond

PHASE 0

Feasibility

- Channel and link model
- Pick the architecture
- Freeze block specs

PHASE 1

Building blocks

- HBT / CML cell library
- Driver, CTLE, sampler
- Test-chip shuttle

PHASE 2

5 Gb/s transceiver

- TX + RX + PLL + CDR
- On-chip PRBS loopback
- Full-chip tapeout

First silicon

PHASE 3

Measure & publish

- Test PCB, eye and BER
- Open measurement data
- Paper: JSSC / TCAS, SMACD 2027

PHASE 4

Beyond

- 10 Gb/s, adaptive DFE
- PAM4 signalling
- Multi-lane and chiplet links

Every phase ends with a tagged release: **design files, models and data, all open.**

Get involved

System modeling
Python / Julia link models, channels, the GUI

Analog & RF
CML drivers, CTLEs, samplers, PLL and VCO

Digital
SystemVerilog SER/DES, PRBS, CDR and DFE logic

Layout & EM
KLayout, Magic, openEMS, parasitic extraction

Test & measurement
Test PCBs, eye diagrams, BER

Docs & teaching
Tutorials, Quarto pages, GitHub docs

Interested? **bricd.org@gmail.com**

`[GitHub repo link]`

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